Huawei Access Node JAY-ACC-68 Optical Transport Failure Standard Operating Procedure (SOP)

Node: JAY-ACC-68 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_7250

Description

This SOP outlines the procedures for diagnosing and resolving Optical Transport Failures on the Huawei Access Node JAY-ACC-68 at the Jayanagar Metro Site. The failure domain is primarily attributed to SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The Access Node is equipped with Huawei's NE40E series, which employs a sophisticated Optical Transport Network (OTN) architecture. The OTN protocol stack consists of multiple layers, including the Optical Transport Unit (OTU), Optical Channel (OCh), and Optical Transport Section (OTS). Each layer is responsible for transporting the optical signal across the fiber network, with the OTU layer being responsible for framing and multiplexing the signal. The failure of any of these layers can result in a loss of optical signal quality, leading to the ALARM_7250 error code. This SOP aims to identify and rectify the root cause of the failure, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Ensure remote hands are dispatched to the Jayanagar Metro Site to provide physical access to the NE40E Access Node.

- Verify the status of the redundant link to ensure it is operational and can take over in case of a failure.

- Check the environmental conditions at the site, including temperature, humidity, and power supply, to rule out any external factors contributing to the failure.

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: display optical-fiber link state, Expected Output: Optical fiber link state: Down, Description: This command displays the current state of the optical fiber link, indicating whether it is up or down. A down state may indicate a physical issue with the fiber or a problem with the SFP/XFP.

Command: display optical-transport-unit otu-status, Expected Output: OTU status: Down, Description: This command displays the OTU status, which may indicate a framing or multiplexing issue. A down status may suggest a problem with the OTU layer.
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```
Command: display optical-channel och-status, Expected Output: OCH status: Down, Description: This command displays the OCH status, which may indicate a signal quality issue. A down status may suggest a problem with the OCH layer.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace sfp xfp module , Description: Replace the SFP/XFP module in the specified slot to resolve the issue. Verify the new module is properly seated and configured before proceeding.

Step 2: Step: Verify fiber integrity, Command: verify fiber integrity , Description: Verify the fiber integrity by checking for any physical damage or cuts. If damage is found, replace the fiber and re-test the link.

Step 3: Step: Adjust Rx/Tx power levels, Command: adjust rx-tx power levels , Description: Adjust the Rx/Tx power levels on the specified port to optimize signal quality and prevent power loss.

Post-Check Verification

```
Command: display optical-fiber link state, Expected Output: Optical fiber link state: Up, Description: Verify the optical fiber link is up and functioning correctly.

Command: display optical-transport-unit otu-status, Expected Output: OTU status: Up, Description: Verify the OTU status is up and functioning correctly.

Command: display optical-channel och-status, Expected Output: OCH status: Up, Description: Verify the OCH status is up and functioning correctly.
```

Roll-back Procedures

- Step: Restore original SFP/XFP module, Command: restore original sfp xfp module , Description: Restore the original SFP/XFP module in the specified slot if the replacement fails to resolve the issue.

- Step: Re-verify fiber integrity, Command: re-verify fiber integrity , Description: Re-verify the fiber integrity by checking for any physical damage or cuts. If damage is found, replace the fiber and re-test the link.

- Step: Re-adjust Rx/Tx power levels, Command: re-adjust rx-tx power levels , Description: Re-adjust the Rx/Tx power levels on the specified port to revert to the original settings.